LEARNER GUIDE

A line is a set of solutions

Equations, Inequalities and Simultaneous Relationships · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Every equation so far had one unknown and (usually) one answer. An equation with **two** unknowns — like $x + y = 6$, or $y = 2x + 1$ — is different: many pairs (x, y) make it true. (0, 6), (1, 5) and (2, 4) all satisfy $x + y = 6$. A **solution** is now a **pair**, and if you plot every pair that works, they line up exactly: the equation's solutions *are* a straight line. Every point on the line is a solution; every solution is a point on the line. So you can test a point by substituting, and read a new solution straight off the line.

What you are learning to notice

- With two unknowns, a solution is a **pair** (x, y), not a single number.
- All the pairs that satisfy the equation fall on one straight line.
- A point is a solution *exactly when* it lies on the line — substitute to check.
- You can read a solution off the line both ways: give an x, get a y; give a y, get an x.

Moving through it, one step at a time

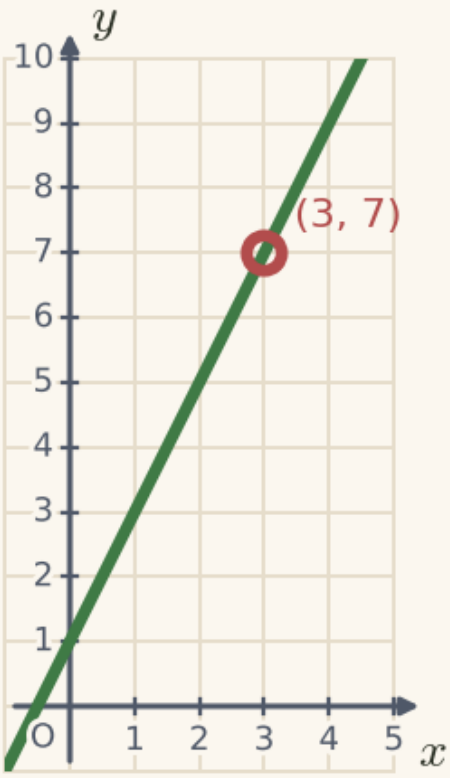
To see the solution set, make a table: choose several x values, work out $y = mx + c$ for each, and plot the pairs — they form the line. To test whether a point is a solution, substitute its x and y into the equation; if both sides match, it is on the line. To read a solution off, pick an x and compute its y (or pick a y and solve for x). Each pair is one solution chosen from the infinitely many on the line.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Is this point a solution?

Is (3, 7) a solution of $y = 2x + 1$?

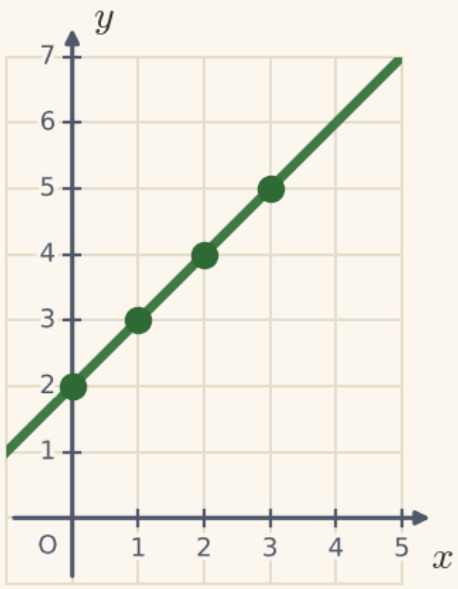


1. Substitute the point's x into the right-hand side.	$2 \times 3 + 1 = 7$
2. Compare with the point's y.	$7 = 7$ ✓
3. Both sides match, so the point is on the line.	(3, 7) is a solution

Yes — 2×3 + 1 = 7 equals the y-coordinate, so (3, 7) lies on the line and is a solution. (3, 5) would give 7 ≠ 5, so it is not.)

Example 2 — A table becomes the line

Plot solutions of $y = x + 2$ from a table.



1. Choose $x = 0, 1, 2, 3$ and work out $y = x + 2$.	$y = 2, 3, 4, 5$
2. Plot the pairs (0,2), (1,3), (2,4), (3,5).	four points
3. They fall on one line — the set of all solutions.	$y = x + 2$

The four pairs lie on one straight line; the table and the line are the same set of solutions, shown two ways.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (coordinates, evaluating $y = mx + c$, and substituting a point into an equation), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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